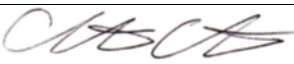


Guideline Title: Overdose, Poisoning, & Toxic Exposure

Guideline Number: GUID-3203-M019

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To recognize the patient with an ingestion or exposure to a toxic substance and to provide and maintain adequate oxygenation/ventilation and tissue perfusion. To provide emergent treatment and expeditious transport to the nearest appropriate medical facility. Toxic ingestion or exposure should be considered in the differential diagnosis of any unexplained symptoms or signs.
- II. **DEFINITIONS:**
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by *GUID3203-G002 – General Patient Care*.
 - b. Attempt to obtain exact time of ingestion, amount and type of ingestion
 - c. Ensure the safety of emergency responders.
 - d. Have all patients decontaminated prior to treatment, as necessary.
 - i. If eye or skin contamination: irrigate with copious amounts of NS.9% or tap water
 - ii. Dry chemicals should be brushed off prior to decontamination.
 - e. Treatment is supportive unless a specific toxicologic symptom complex is diagnosed.
 - f. Treat dysrhythmias according to *GUID3203-C001-C010 - Cardiac Guidelines*.
 - g. Verify blood glucose check and if glucose is < 60 mg/dL and the patient is symptomatic, treat per *GUID3203-M021 – Hypoglycemia*.
 - h. If patient displays signs of inadequate perfusion treat per *GUID3203-M011 – Hypotension*.
 - i. Treat seizures per *GUID3203-M015 – Seizures*.
 - j. For suspected opiate overdose
 - i. Adult & Pediatric > 20 kg: **Naloxone 0.4 to 2 mg IV/IM or 2 mg IN**. Repeat q 3-5 minutes. Fentanyl overdoses may require high total doses (8-10 mg)
 - ii. Pediatric <20kg: **Naloxone 0.01-0.1 mg/kg IV/IN**. Maximum single dose 2mg.
 - k. Consider contacting local **Poison Control Center 1-800-222-1222**. Administer known antidote per direction of Medical Control or Poison Control Center.
 - l. Consider sedation if agitated per the *GUID3203-G006 Management for the Anxiety/ Combative Patient*.
 - m. See chart below for specific management and discuss with sending/receiving physician.
- IV. **DOCUMENTATION:**

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a. EMS Charts

V. REFERENCES:

Specific Drug Toxicities

Drug or Poison	Source	Signs/Symptoms	Management with physician order
Acetaminophen Ingestion of 150 mg/kg or 7.5 g is potentially toxic	Tylenol, (Comtrex, Contact, Darvocet-N, Nyquil, Sinutab)	- Initially little to no S & S - Malaise, nausea, & vomiting. - Late signs: abdominal pain & signs of hepatic injury/necroses.	- Serum levels: draw 4 hours after ingestion. - N-Acetylcysteine (Mucomyst) 150 mg/kg loading followed by 50 mg/kg next 4hrs IV
ASA Toxic dose 200-300 mg/kg, 500 mg/kg potentially lethal	Aspirin, many OTC analgesics, oil of wintergreen	- Gastritis, Respiratory alkalosis, Metabolic acidosis, ARDS, Late CNS depression	- Activated charcoal 1 gram/kg PO up to 50 gms - Treat dehydration, replace K+ - Hemodialysis if decreased LOC, pulmonary manifestations, and acid base imbalance - Use caution w/ tracheal intubation & reserve for respiratory failure.
Anticholinergics	Antihistamines, TCA's, Allerest, phenothiazines, Sominex, Contac,	- Agitation, anxiety, and hallucinations, Dilated pupils, Seizures, Tachycardia, Decreased secretions	- Symptomatic therapy - Physostigmine 1-2 mg IV over 20 min very short acting total dose of 6mg can be given - Treat cardiac arrhythmias
β-Blockers	Lopressor, Propanolol	- Bradycardia with AVB, Hypotension, Decreased LOC, Hypoglycemia, Seizure, Pulmonary edema	- Supportive care - For bradycardia: Atropine 0.5-1mg IV - For hypotension: Epinephrine 0.1-0.5 mcg/kg/min & Glucagon 0.1 mg/kg IV (max 10mg) load followed by 2-5 mg/hr infusion. Peds 0.1mg/kg IV. - Consider high dose insulin therapy (1 unit/kg gtt) with glucose supplementation as needed to keep BGL > 100 - Continue Lipid Emulsion Therapy 0.25-0.5 mL/kg per hour if began by sending facility
Benzodiazepines	Valium, Librium, Dalmane	- Drowsiness, Ataxia, Nystagmus, Slurred speech, Respiratory depression, Hypotension	-Supportive care. Flumazenil contraindicated for benzo overdose due to risk of seizures
Calcium Channel Blockers	Amlodipine/Norvasc, Diltiazem/Cardizem, Tiazac, Felodipine, Isradipine, Nicardipine, Nifedipine/Procardia, Nisoldipine/Sular, Verapamil	- Bradycardia, Hypotension, Respiratory depression, N/V	- Supportive care - For hypotension: Calcium chloride 1 g IV can repeat in 10 min and glucagon 0.1 mg/kg IV load followed by 2-5 mg/hr infusion - Calcium Gluconate 1 gram or 60mg/kg/dose slow IV over 5 minutes. May repeat every 10 minutes. - Consider high dose insulin therapy (1 unit/kg gtt) with glucose supplementation as needed to keep BGL > 100 - Continue Lipid Emulsion Therapy 0.25-0.5 mL/kg per hour if began by sending facility
Carbon monoxide		- Common: Headache, dizziness, nausea, vomiting, chest pain, weakness, and AMS.	- 100% oxygen via non-rebreather - Intubate if unconscious

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		- Severe: malaise, SOB, HA, nausea, CP, irritability, ataxia, AMS, LOC, coma, death; tachycardia, impaired memory, tachypnea, hypotension, cognitive sensory disturbances; metabolic acidosis, arrhythmias, MI, and pulmonary edema	
Cocaine & Methamphetamine		- Vasoconstriction, tachycardia, HTN, RF, increased myocardial O2 demand, coma, seizures, supraventricular & ventricular dysrhythmias, agitation, HA, intracranial hemorrhage, muscle pain, compartment syndrome, rhabdomyolysis, hyperkalemia	- Large doses of benzodiazepines - Beta-blockers contraindicated - Avoid succinylcholine
Digoxin	Lanoxicaps, Lanoxin,	- Nausea, vomiting, hyperkalemia, brady dysrhythmias HR<60 or atrioventricular block, tachydysrhythmias V-tach V-fib, atrial tachycardia with 2:1 block	- Digibind is antidote - Watch patient for hyperkalemia - Do not give Calcium chloride or calcium gluconate
Ethylene Glycol As little as 2ml/kg is fatal dose	Antifreeze, coolant	- CNS depression, Vomiting, Abdominal pain, Late high anion gap acidosis, Seizure, Coma, RF, calcium oxalate crystals in urine	- Supportive care - Ethanol infusion start at 1000 mg/kg/hr for 1-2 hrs then 100 mg/kg/hr to keep ethanol level at 100-130 mg/dL
Organophosphates		- Muscarinic effects: bradycardia, bronch-orrhea, respiratory distress, visual disturbance, and SLUDGE syndrome (Salivation, Lacrimation, urination, defecation, gastrointestinal, and emesis) - Nicotinic effects: tachycardia, muscle fasciculation with weakness or paralysis, hypertension - CNS effects- headache, AMS, anxiety, lethargy, coma	- Atropine 2mg or 0.5mg/kg IV , repeated every 2 minutes until reversal of symptoms - Pralidoxime (2-PAM) 600mg (or 1 auto-injector IM) or 20-50mg/kg IV for pediatric patients. May repeat x2 to total of 1,800mg.
Opiates	Dilaudid, fentanyl, methadone, Demerol	- Pinpoint pupils, Respiratory depression, Coma, ARDS	- Supportive care - Narcan 0.4-2 mg IV/IN , may need infusion (0.4-0.8 mg/hr to max of 1.2 mg/hr)
Sodium channel blocker Tricyclic antidepressant (TCA)	Amitriptyline, doxepin, imipramine, protriptyline	- Anticholinergic symptoms, Delirium to coma, Dilated pupils, Dry membranes, SVT, Conduction disturbances, Prolonged QT, Hypotension	- Serious toxicity usually within 6 hrs of ingestion - If arrhythmias & hypotension: Sodium Bicarb 50 mEq in 250ml D5W IV over 1 hour may require infusion (keep pH @ 7.45-7.55) - Hypotension: IV fluid bolus then Norepinephrine or Epinephrine infusion - Benzodiazepines for seizures